

# Viewpoints: Impact of Big Data on Data Engineering

Can you talk about how big data is influencing and shaping the field of data engineering?



## Impact of Big Data on Data Engineering

- Big Data can be described as data that has the attributes of Velocity, Veracity, Volume, and Variety – the four Vs



## Impact of Big Data on Data Engineering

- Big Data has had a major impact on Data Engineering, making it a much more diverse and rich field
- It is critical for organizations, today, to be able to derive insights from the large volumes of data they are collecting
- Emergence of new technologies and products related to working with Big Data



## Impact of Big Data on Data Engineering

- Created a huge demand for professionals who can work with Big Data for building and managing Big Data systems and analyzing Big Data



## Impact of Big Data on Data Engineering

- In the pre-IoT and social media days, the number of ways in which you could ingest data into a database were limited



## Impact of Big Data on Data Engineering

- In the past decade, with gadgets and devices talking to each other, the nature of data has changed
- As a result of the evolution of data, data engineering too has evolved
- Traditional RDBMSes are not a one-size fit-all solution
- Newer technologies - Google BigTable, Cassandra, Graph-based Databases, Hadoop, MapReduce



## Impact of Big Data on Data Engineering

- With Big Data, organizations have become more accepting of storing large amounts of data
- Big Data has led to better ways of handling large amounts of data
- Big Data has also led to better handling of large amounts of unstructured data



## Conclusion

Big Data refers to the vast amounts of data that is being produced each moment of every day, by people, tools, and machines. The sheer velocity, volume, and variety of data challenged the tools and systems used for conventional data, leading to the emergence of processing tools and platforms designed specifically for Big Data.

Big Data processing technologies help derive value from big data. These include NoSQL databases and Data Lakes, and open-source technologies such as Apache Hadoop, Apache Hive, and Apache Spark.

- Hadoop provides distributed storage and processing of large datasets across clusters of computers. One of its main components, the Hadoop File Distribution System, or HDFS, is a storage system for big data.
- Hive is a data warehouse software for reading, writing, and managing large datasets.
- Spark is a general-purpose data processing engine designed to extract and process large volumes of data.